The Difference Field and Emergent Invariants: A Conceptual Note on the Subject*

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This article proposes a conceptual framework in which the world accessible to the subject (S) is not identified with the fullness of reality but is considered as a system of invariant structures ($\mathcal{I}$) available for perception and coordination. The starting point is a limiting thought experiment in which the subject (S), placed in a nearly homogeneous environment physically saturated with data, is deprived of accessible external supports for stable discrimination. On this basis, the notion of the Difference Field (D) is introduced as a pre-object foundation that coincides neither with the ready-made world of objects nor with information for a particular subject (S).

It is shown that information for the subject (S) arises only where data become available in the form of invariant structures ($\mathcal{I}$), while space and time may be interpreted as derivative modes of coordinating such structures rather than as unconditionally given preconditions of experience. The article further outlines a hypothesis of multilevel compression, within which the subject (S) is regarded as a higher-order invariant structure that does not cognize the Difference Field (D) in its fullness but coordinates the accessible part of the field while preserving its own integrity.

In this sense, the concept of the subject S is not reduced to a mere point of observation but approaches the idea of an integral model of selfhood, albeit within a different conceptual language [1].

The article has the character of a first conceptual presentation of the proposed idea and is intended as an invitation to critical interdisciplinary discussion.

I. INTRODUCTION

The conventional scientific description of the world usually proceeds from the assumption that there are objects situated in space and time, while the subject (S) merely registers or increasingly accurately describes an already existing structure. Yet such a picture may be not primary but derivative. What appears to us as a ready-made world of things, events, distances, and durations may instead be the result of a certain mode of stabilizing differences accessible to a particular type of subject (S).

This article proceeds from the assumption that, in order to construct a more fundamental framework, one should distinguish at least three levels: that which exists independently of a particular subject (S); that which can be stably distinguished in the form of invariants (I) or systems of invariants ($\mathcal{I}$); and that which becomes a coordinated world for the subject (S). This distinction is important because in both everyday and scientific thinking these levels usually merge without explicit awareness. We tend to identify reality with the world available to us, and the world with a set of already formed objects. Yet there is no reason to assume that any possible subject (S) would have to see the same world in exactly the same structure.

To make this problem as clear as possible, the article considers a limiting thought experiment in which the subject (S) is placed in a nearly homogeneous environment where a large amount of data is physically present but accessible external invariants (I) are almost absent. The imaginary character of this example is partly inspired by a perceptual situation close to that created by Antony Gormley's installation *Blind Light* (2007). The purpose of this experiment is not to empirically prove a new theory, but to reveal the limit of the intuition according to which the world is automatically thought as already structured.

The main thesis of this work is that the presence of data does not yet guarantee the presence of a world for the subject (S). A world arises only where data can be stably distinguished, compared, and integrated into a coordinated system of invariants ($\mathcal{I}$). On this basis, the notion of the Difference Field (D) is introduced in the article and is regarded as a pre-object foundation that coincides neither with information for a particular subject (S) nor with already constructed space and time. The article then advances the hypothesis that the subject (S) may be interpreted as a higher-order invariant (I) that does not encompass the Difference Field (D) in its fullness but coordinates its accessible part while preserving the possibility of experiencing itself as an integral, though invariant, structure.

The article does not claim the status of a completed physical or mathematical theory. Its aim is more modest: to propose a conceptual language that allows the Difference Field D, invariant structures I, multilevel compression,

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and the subject S to be considered together. Methodologically, the proposed framework is close to the systems tradition in which reality is not reduced to a set of separate things but is regarded as a field of relations, differences, and levels of organization from which stable forms of the world emerge [2]. Within this perspective, the subject S may be interpreted as a higher-order invariant structure that maintains a coordinated accessible world through the selection, compression, and stabilization of invariant structures I. This interpretation, in turn, partially resonates with contemporary models in which a cognitive system preserves its integrity by minimizing uncertainty [3].

II. LIMITING THOUGHT EXPERIMENT: A ROOM FILLED WITH HOMOGENEOUS FOG

Let us imagine a room completely filled with dense, dry, uniformly illuminated fog. The light in it is uniform: there is no visible boundary, source of light, direction, or contrast. The subject S is located at the center of this room, does not touch the walls, and cannot see its own body. Its arms are lowered, and the fog is so dense that the subject S does not even register the presence of its own nose in current perceptual experience. The environment is so homogeneous that any local features disappear into the general background.

Physically, such a room is by no means empty. It contains a large number of water molecules, photons, local inhomogeneities, and possible configurations. Theoretically, one could select a circle from certain molecules and another circle of smaller size from others; comparing them could generate a relation analogous to a spatial one. Without changing the positions of the molecules, but merely selecting them in a certain order, one could in principle reconstruct arbitrarily complex structures—for example, the full text of the *Encyclopaedia Britannica*. Yet for the subject (S) located in the room, these possibilities remain inaccessible. It does not distinguish individual molecules, does not see their mutual relations, and has no means of stably isolating any of these potential structures.

As a result, a peculiar experience arises: the surrounding environment is physically saturated, yet there is almost no world as a system of accessible objects and relations. The subject (S) cannot reliably isolate any external point to which a coordinate system could be anchored. It has no sufficient basis for constructing even the simplest geometry. If one further assumes the absence of a stable gravitational orientation, for example in a state of uniform fall, then even the last external direction that might have been used as a point of reference disappears.

Despite this, a complete absence of experience does not occur. The subject (S) retains internal rhythms: heartbeat, breathing, residual bodily sensitivity. These become almost the only accessible invariant structures ($\mathcal{I}$). Through them, a minimal experience of one's own continuity is preserved even if the external world almost disappears.

It is especially important that, in such a situation, the very notion of time becomes problematic. If external events are not isolated and changes cannot be stably correlated, then the subject (S) has no basis from which to construct a sequence. Without accessible cycles and recurrent differences, changes do not become events, and without events time in the usual sense does not arise. The only basis for the experience of duration remains the body's internal rhythms. This has an obvious affinity with embodied cognition approaches [4].

Similarly, space also ceases to be something unconditionally given. Space in the usual sense requires at least two stably correlated invariant structures ($\mathcal{I}$) between which one can establish a relation of distance, direction, or boundary. In a room filled with homogeneous fog, such structures are almost absent. Thus space does not disappear as a physical possibility, but ceases to exist as an accessible form of the organization of experience for the particular subject (S) present in this room.

This thought experiment makes it possible to draw a first principled conclusion: the presence of physical data does not yet guarantee the presence of a world for the subject (S). A world arises only where there are invariant structures ($\mathcal{I}$) that can be stably distinguished, repeatedly correlated, and integrated into a coordinated picture for the subject (S). This difference between the physical saturation of the environment and the absence of an accessible world is precisely the point of departure for introducing the concept of the Difference Field (D).

III. DATA, INFORMATION, AND THE ACCESSIBLE WORLD

The example of the room shows that there is a principled difference between the physical presence of an environment and the world experienced by the subject (S). To make this difference explicit, it is useful to distinguish at least three notions: data, information, and the accessible world.

By data, this article means the entire totality of differences, configurations, and potential relations that are physically present regardless of whether a particular subject (S) is able to distinguish them. In the room filled with fog, data are extraordinarily abundant: molecules, photons, local inhomogeneities, possible geometric configurations. Yet the mere presence of data does not mean that they have significance for the subject (S).

Information arises only when certain data can be distinguished as invariant structures ($\mathcal{I}$), that is, as forms that preserve sufficient stability for repeated discrimination and correlation. If the subject (S) has no way of stably isolating

a given structure, then for that subject it is not information, even if it is physically present. In the room with fog, the subject (S) is theoretically surrounded by a vast number of possible structures, yet almost none of them becomes information because it cannot be repeatedly recognized by the subject (S).

This allows the following definition to be formulated:

> Information for the subject (S) is data that have become accessible in the form of invariant structures ($\mathcal{I}$) suitable for repeated correlation.

This definition differs significantly from approaches in which information is treated as an objective property of any physical configuration. Within the proposed framework, one and the same configuration can be information for one type of subject (S) and not be information for another. Information is not simply "that which is"; it is that which can be invariantly accessible.

The accessible world, in turn, is not the totality of information, but a coordinated system of invariant structures ($\mathcal{I}$) that can be integrated into a stable picture. For a human being under ordinary conditions, the accessible world includes spatial objects, boundaries, motion, causality, and temporal sequence. In the room with homogeneous fog, this world almost collapses: most familiar external invariant structures ($\mathcal{I}$) become inaccessible, and therefore the external world ceases to be organized.

Yet even in this situation the accessible world does not disappear completely. The body's internal rhythms—heartbeat, breathing, changes in bodily tension—remain accessible invariant structures ($\mathcal{I}$). They constitute the minimal remainder of a world for the subject (S). Thus the accessible world is not identical to the external environment; it is determined by which invariant structures ($\mathcal{I}$) the subject (S) is capable of maintaining and integrating.

Thus, there is a hierarchy between data, information, and world:

$$\text{data} \supset \text{information for } S \supset \text{accessible world}$$

Data encompass everything physically possible. Information constitutes only that part of the data that has become accessible as an invariant structure. The accessible world is narrower still: it is not merely a set of separate invariants (I), but their coordinated system ($\mathcal{I}$).

This hierarchy explains why the subject (S) in the room experiences something close to the absence of a world, although physically it is located in an environment extraordinarily rich in data. What is experienced as "nothing" is not the absence of physical reality, but the absence of accessible invariant structures ($\mathcal{I}$) sufficient for the construction of a world. The next step is to introduce the concept of the Difference Field (D), which allows data to be thought of as a more fundamental level than any already formed world.

IV. THE DIFFERENCE FIELD (D)

If data do not coincide with information for the subject (S), and information does not coincide with the accessible world, then the question arises: what exactly is that level at which data exist prior to any distinction, interpretation, or coordination? In this article, this level is proposed to be designated as the Difference Field (D).

By the Difference Field (D) is meant the most general totality of all possible differences and relations that have not yet been organized into a world of objects, events, coordinates, or meanings. (D) is not a set of already given things. Within it there are as yet no apples, bodies, particles, directions, or moments of time. Nor should (D) be identified with information: information arises only when some part of (D) becomes accessible as an invariant structure for a particular subject (S).

In this sense, the field (D) is a pre-object foundation. It does not contain space and time as already given forms. Space and time may arise only later, as a result of the coordination of invariant structures ($\mathcal{I}$). What in ordinary experience appears as three-dimensional space or a sequence of events is, within the proposed framework, not a primary property of (D), but only one of its possible modes of organization.

It is also important that (D) is not emptiness. On the contrary, the room filled with homogeneous fog shows that a situation is possible in which the subject (S) experiences an almost complete absence of a world while an excess of physical data remains all around. Therefore, the experience of "nothing" should not be interpreted as the absence of the field (D). It only means that for this particular subject (S) under these conditions, accessible invariant structures ($\mathcal{I}$) are almost absent.

The field (D) should also not be thought of as a homogeneous mass without internal organization. Even in the room with fog it is theoretically possible to isolate certain configurations from the molecules: for example, a circle or another form. But these configurations are not already given invariant structures. They exist only as a potential possibility. Precisely for this reason, (D) can be described as a field not of ready-made objects but of potential differentiability.

Within this article, it is proposed to proceed from the assumption that in the field (D) there do not exist two absolutely identical elements or configurations. Identity arises only at the level of further coordination, when certain differences begin to be compressed into a single invariant structure (I). What appears to the subject (S) as "the same" may remain, at the level of (D), a set of distinct yet sufficiently close configurations.

Thus, sameness is not a fundamental property of reality but a result of the compression of differences. It is precisely thanks to this that repetition, objects, and stable worlds become possible later.

The field (D) is not a world for any particular subject (S). No subject (S) can fully encompass (D) or retain all its differences within the bounds of its own experience. The reason lies not only in the limitation of the subject, but also in the very nature of (S): in order to be an integral invariant structure, the subject (S) is compelled to compress, discard, and order most of (D). Thus, any world accessible to the subject (S) is only a local and partial manifestation of the Difference Field.

This makes it possible to formulate the central thesis:

> (D) is not the world; the world is an invariantly accessible manifestation of (D) for a particular subject (S).

The next step is to explain how invariant structures ($\mathcal{I}$) can arise from the field (D) and why it is precisely they that become the minimal condition for the emergence of a world for the subject (S).

V. INVARIANT STRUCTURES AS THE CONDITION OF A WORLD

If the field (D) itself is not yet a world, then the question arises: what exactly must emerge for a world to become possible for the subject (S)? Within the proposed framework, the minimal condition for this is the emergence of invariant structures ($\mathcal{I}$).

By an invariant structure, this article means a form that can be repeatedly isolated and correlated despite local changes or variations. An invariant structure (I) need not be absolutely unchanging. On the contrary, it may consist of many distinct configurations that, at a higher level of compression, are perceived as "the same."

For example, what a person calls one and the same heartbeat never repeats absolutely exactly. Each heartbeat differs from the previous one. Yet this sequence can be compressed into a single invariant structure (I)—"heartbeat." Similarly, in everyday experience the subject (S) perceives "the same" object even when lighting, position, angle of view, or minor details change.

Thus, an invariant structure (I) is not identity in the strict sense. It is the result of compressing a multiplicity of differences into a form sufficiently stable for repeated correlation.

This makes it possible to provide the following definition:

> An invariant structure (I) is the compression of a multiplicity of differences from the field (D) into a form that can be repeatedly recognized and correlated.

It is precisely invariant structures ($\mathcal{I}$) that make the emergence of a world possible. Without them, the subject (S) cannot distinguish "the same" from "the other," cannot construct a boundary, an object, or a sequence. In the room with homogeneous fog, the external world nearly disappears precisely because the environment does not provide enough accessible invariant structures ($\mathcal{I}$). The subject (S) cannot stably isolate direction, boundary, or form.

Despite this, some invariant structures ($\mathcal{I}$) still remain. Above all, these are the internal rhythms of the organism: heartbeat, breathing, bodily tension. They are minimal structures that the subject (S) can repeatedly correlate. This is why, even in a situation of almost complete loss of the external world, the experience of one's own continuity does not disappear entirely.

Within this article it is useful to distinguish two types of invariant structures:

1. **External invariant structures** — those connected with relatively stable features of the environment;
2. **Internal invariant structures** — those connected with the subject (S) itself and supporting its minimal integrity.

In ordinary human experience, these two types are closely intertwined. The external world is constantly correlated with internal rhythms and states, while internal integrity is maintained through interaction with external invariant structures ($\mathcal{I}$). In the room with fog, this balance is radically shifted: almost all external invariant structures disappear, and the subject (S) is left almost alone with its own internal invariants.

It is especially important that invariant structures ($\mathcal{I}$) not only allow the subject (S) to see a world, but also determine which world is possible for it. Another type of subject (S) that had access to different invariant structures

($\mathcal{I}$) could construct an entirely different picture of reality. What for a human is an object, space, or event may not exist for another type of subject (S), and vice versa.

From this it follows that the world is not a simple aggregate of things equally accessible to all possible subjects (S). The world is a system of invariant structures ($\mathcal{I}$) accessible to a particular type of subject (S).

That is why invariant structures ($\mathcal{I}$) should not be regarded as an addition to an already given world. They are the condition of its emergence. Without them there exists only the field (D), saturated with differences but not yet organized into a world.

VI. SPACE AND TIME AS DERIVATIVE MODES OF COORDINATION

In ordinary experience, space and time appear to be the most fundamental characteristics of the world. We are inclined to assume that objects are always already located in space and events are always already occurring in time. Yet the example of the room shows that this intuition may be derivative.

In the room with homogeneous fog, the environment is physically present, but the subject (S) does not possess enough accessible invariant structures ($\mathcal{I}$) to construct even the simplest geometry. Space in the usual sense requires at least two stably correlated invariant structures (I) between which one can establish a relation: nearer or farther, higher or lower, left or right. If such structures are absent, then there is no basis for the emergence of coordinates, boundaries, or directions.

Thus, space is not simply a "container" that exists independently of any mode of perception. It arises only where invariant structures ($\mathcal{I}$) can be stably correlated with one another. Space, in this sense, is a mode of organization of accessible differences rather than a primary property of the field (D).

Similarly, time also cannot be derived directly from the mere fact of change. For time to arise, it is not enough that something occurs. It is necessary that different states be correlated as repetition, sequence, or cycle. If the subject (S) does not have access to invariant structures ($\mathcal{I}$) that can be compared, then changes do not become events, and without events time in the usual sense does not arise.

In the room with fog this is especially evident. External changes, even if they exist, can hardly be integrated into a sequence. Therefore the only source of the experience of duration becomes internal invariant structures ($\mathcal{I}$): heartbeat, breathing, changes in bodily state. It is precisely these that create the minimal basis for what the subject (S) can experience as "before" and "after."

From this it follows that time is not an independent substance or background within which events occur. Time arises as a result of the coordination of invariant structures ($\mathcal{I}$) that can be correlated in a sequence.

Within the proposed framework, the following definitions may be formulated:

> Space is a mode of coordinating coexisting invariant structures ($\mathcal{I}$).
>
> Time is a mode of coordinating successive invariant structures ($\mathcal{I}$).

Both of these modes do not exist at the level of the field (D) as such. They arise only when some part of (D) becomes accessible in the form of invariant structures ($\mathcal{I}$) and can be integrated into a stable picture.

Therefore, space and time are not primary preconditions of the world. On the contrary, they themselves are derivative of a more fundamental process—the emergence, correlation, and compression of invariant structures ($\mathcal{I}$). That is why different types of subject (S), which have access to different invariant structures ($\mathcal{I}$), could potentially construct entirely different forms of space and time, or none at all.

The example of the room demonstrates this in the simplest way: the physical environment remains, but space and time almost disappear as accessible forms of the organization of experience. This means that they cannot be accepted as unconditionally given properties of reality. They are only one of the possible ways in which a certain type of subject (S) organizes the part of the field (D) accessible to it.

VII. MULTILEVEL COMPRESSION

If the field (D) is not an already given world, and invariant structures ($\mathcal{I}$) are the condition for the emergence of a world for the subject (S), then the question arises: how can such structures emerge from the field of differences? Within this article, the hypothesis is advanced that between (D) and the accessible world there exists a sequence of levels of compression.

Let (D) be the primary field of differences in which there are as yet neither space nor time nor ready-made invariant structures ($\mathcal{I}$). Then the first compression gives rise to a new level, which we shall denote as (D'). (D') is not merely

a subset of (D). It is a new field that arises as a result of compressing part of the differences of (D) into forms suitable for repeated correlation.

Schematically, this can be written as:

$$C(D) = D'$$

where (C) denotes the compression operator.

The operator (C) does not eliminate differences completely. Its action consists in the fact that a multiplicity of different configurations begins to be treated as a single invariant structure (I). What exists at the level of (D) as a multitude of distinct configurations may, at the level of (D'), be compressed into one form that becomes accessible for repeated correlation.

It is fundamentally important that (D) does not disappear in the process. Compression is not the destruction of the previous level, but only the emergence of a new level of organization. Therefore (D) and (D') coexist.

Further acts of compression may generate new levels:

$$D \to D' \to D'' \to D''' \to \dots$$

Each new level is characterized by a lower descriptive capacity. What at a lower level appears as a vast number of differences may, at a higher level, be compressed into a small number of invariant structures ($\mathcal{I}$).

The idea of multilevel compression, in which each subsequent level decreases diversity while increasing the stability and integrability of forms, partly resonates with notions of multiscale organization and self-similarity [5, 6].

Thus, each act of compression simultaneously:

- reduces the number of accessible differences;
- increases the stability of invariant structures I;
- makes possible a new level of coordination.

It is precisely thanks to this that the emergence of ever more complex invariant structures ($\mathcal{I}$) becomes possible. The level (D') may contain the simplest structures of repetition. The level (D") may contain their more stable combinations. At yet higher levels there may arise structures capable not only of maintaining their own integrity but also of correlating themselves with other structures.

It is important that each new level rests only on part of the previous one. Not all differences of (D) enter into (D'), and not all structures of (D') pass into (D"). Therefore every act of compression is simultaneously an act of selection and loss. It does not open up the whole possible world, but only a certain mode of its organization.

It is precisely here that an explanation becomes possible of why different types of subject (S) can see different worlds. If a certain type of subject (S) arises on the basis of a specific sequence of compressions, then this sequence determines which invariant structures ($\mathcal{I}$) are accessible to it and which remain invisible. The world of *Homo sapiens* is only one of the variants of compression.

At the same time, different sequences of compression may partially overlap. Therefore different subjects (S) can share elements of a world even if their invariant structures ($\mathcal{I}$) do not fully coincide. This may explain why people, despite having different individual experiences, still see approximately the same world.

It is especially important that lower levels do not cease to affect higher ones. Even if some structure has already become invariant at the level of (D") or (D"'), it continues to depend on lower levels at which significantly more differences remain. This means that connections may be preserved between different invariant structures of the higher level through a shared lower basis. In future work, such multilevel interaction may become the basis for interpreting physical fields, nonlocal effects, or action at a distance.

Within this article it is sufficient to formulate the main principle:

> The world accessible to the subject (S) is not a direct image of the field (D), but the result of successive multilevel compression in which each level decreases diversity and increases invariance.

The concept of the subject (S), introduced intuitively at the beginning of the article, can now be described more precisely as a higher-order invariant structure.

VIII. THE SUBJECT AS A COORDINATED INVARIANT STRUCTURE

Within the proposed framework, the subject (S) cannot be regarded as something primary and self-sufficient. If a world arises only where invariant structures ($\mathcal{I}$) are distinguished and coordinated from the field (D), then (S) itself must also be the result of such compression.

This means that (S) does not precede invariant structures ($\mathcal{I}$), but arises together with them. At the same time, (S) is not merely one invariant structure among others.

In light of this, (S) may be defined as:

> **Definition.** The subject (S) is a higher-order invariant structure (I) that integrates accessible invariant structures ($\mathcal{I}$) into a coordinated world while maintaining the invariance of its own internal model of the world.

Such a definition means that (S) never encompasses the whole field (D). On the contrary, it always arises only on the basis of a certain sequence of compressions and only of a certain set of invariant structures ($\mathcal{I}$). That which does not enter this sequence does not disappear, but does not become part of the accessible world for this subject (S).

Thus any subject (S) is local and partial. It does not know (D) in its fullness, but only constructs a stable model of that part of (D) which has become accessible to it. That is why different subjects (S) can have different worlds even if they arise on the basis of a shared field of differences.

At the same time, subjects (S) of the species *Homo sapiens* usually perceive the world similarly. Within the framework of this article, this can be explained by the fact that subjects (S) of the species *Homo sapiens* arise on the basis of close or partially overlapping sequences of compression. They have similar bodies, similar internal rhythms, similar sensory limitations, and therefore the invariant structures ($\mathcal{I}$) accessible to them largely coincide. It is precisely because of this that people can speak of "the same" world, although no subject (S) has full access to it and the absolute identity of worlds for different subjects is unattainable.

It is especially important that the subject (S) not only integrates the world but also maintains its own invariance. In order to remain itself, the subject (S) is compelled to compress a multitude of differences into a relatively stable form. If this form loses stability, not only the integrity of the subject (S) itself is destroyed, but also the possibility of seeing the world as a coordinated system of invariant structures ($\mathcal{I}$). Within the proposed framework, mental disorders can be interpreted precisely as a partial or complete breakdown of those mechanisms of compression and coordination that support the typical invariance of *Homo sapiens*.

In the room with homogeneous fog this is especially evident. When almost all external invariant structures ($\mathcal{I}$) become inaccessible, the subject (S) does not completely cease to exist. It continues to maintain a minimal form of selfhood through internal invariant structures ($\mathcal{I}$)—heartbeat, breathing, bodily sensation. It is precisely these that support the simplest invariance of (S), even when the external world almost disappears.

At the same time, this thought experiment should not be understood as a neutral or infinitely stable state. Scientific experiments in sensory and perceptual deprivation show that prolonged loss of accessible external invariant structures ($\mathcal{I}$) has a destructive effect on the coordinated subject (S). In studies conducted in the 1950s under the direction of Donald O. Hebb at McGill University, volunteers were placed in conditions of nearly complete sensory deprivation: in small rooms, with their eyes covered, muffled sound, and minimal tactile contact. After only a few hours, many participants experienced difficulties with sequential thinking, and after approximately two days most reported hallucinations, loss of the sense of reality, and breakdown of habitual cognitive structures. Because of the severe psychological impact, the experiments often had to be terminated earlier than planned; almost none of the participants could endure more than a few days [7].

Within the proposed framework, this can be interpreted as a consequence of the fact that internal invariant structures ($\mathcal{I}$) prove insufficient for the long-term maintenance of a coordinated subject (S). When external invariant structures ($\mathcal{I}$) disappear, (S) preserves itself for some time through heartbeat, breathing, and bodily continuity. But if new external invariant structures ($\mathcal{I}$) do not appear, the process of compression begins to lose stability. The coordinated world collapses, and with it the subject (S) itself, as a structure capable of integrating experience, begins to collapse.

Perhaps this is why hallucinations arise under conditions of prolonged deprivation. In this perspective, they can be understood not as a random error of the brain, but as an attempt by the subject (S) to restore absent invariant structures ($\mathcal{I}$) and once again construct at least a minimally coordinated world.

This makes it possible to assert that (S), as a special invariant, is the result not of a single act but of multilevel compression in which increasingly complex invariant structures ($\mathcal{I}$) are integrated into a more stable whole. At lower levels there may exist only separate invariant structures (I). At higher levels there appears the possibility not only of retaining these structures but also of correlating them with one another and preserving the continuity of the very process of correlation. Such a structure is precisely the subject (S).

In this sense, (S) is not a complete image of the world and is not independent of it. The subject (S) and the accessible world arise together as two sides of one and the same process of compression. The world is a system of

invariant structures ($\mathcal{I}$) accessible to (S), while (S) is an invariant structure (I) capable of maintaining this system as coordinated.

IX. LIMITS OF THE HYPOTHESIS AND OPEN QUESTIONS

Although the author treats the framework of the future theory formulated at this stage with considerable caution, already now, on the basis of the considerations presented, it becomes possible to attempt to advance new hypotheses and outline promising research directions in such fields as fundamental physics, mathematics, cybernetics, neurobiology and biology in general, gerontology, the search for mind (not only extraterrestrial), and the development of artificial intelligence. It is precisely for this purpose that the present work is published.

At the same time, the author acknowledges that the proposed framework is only an initial attempt to formalize intuitive reflections on the nature of the world, the subject S, and their interaction. It claims neither completeness nor the status of final truth. On the contrary, there remain within it many open questions, potential weak points, and conceptual tensions that require further analysis.

The high level of abstraction and generalization on which this framework is built may prove to be both its strength and its weakness. On the one hand, it makes it possible to cover a wide range of phenomena and concepts and potentially become a foundation for uniting different scientific directions under a common theoretical perspective. On the other hand, this very level of abstraction may make its verification, formalization, and, accordingly, its acceptance by the scientific community more difficult.

One of the main open questions concerns the status of the Difference Field D itself. For now, it is introduced as a limiting conceptual notion necessary in order not to identify reality with an already formed world of objects. Yet it still remains to be clarified whether D can be strictly defined as a fundamental ontological basis, or whether it should remain only a heuristic instrument of description.

No less important is the question of the nature of invariant structures $\mathcal{I}$. In the article, they are described as the condition for the construction of an accessible world for the subject S, yet it is still unclear according to which criteria a given structure may be considered invariant, where the boundary lies between an individual invariant I and a system of invariants $\mathcal{I}$, and how these structures should be formalized within a future theory.

The concept of compression also requires special attention. In the proposed framework, compression explains the transition from the Difference Field D to invariant structures $\mathcal{I}$, and from them to more complex levels of organization, including the subject S. However, the mechanism of this transition has so far been described only conceptually. It remains necessary to clarify what formal properties the compression operator C must have, under what conditions compression preserves stability, and why some lines of compression can lead to the emergence of a coordinated subject S, whereas others cannot.

Likewise, the question remains open as to how space and time should be understood within this framework. In the article, they are interpreted as derivative modes of coordinating invariant structures $\mathcal{I}$, but this idea does not yet possess a sufficient mathematical apparatus. Future work must show whether it is possible to consistently derive from such a perspective at least some of the known properties of space, time, and causality.

Another important question concerns the status of the subject S itself. In the proposed framework, it is regarded not as a primary entity but as a higher-order invariant structure that maintains its own continuity and coordinates the accessible world. Yet it is still unclear where the boundary lies between a complex system of invariants $\mathcal{I}$ and a subject S, under what conditions such a system acquires selfhood, and how this boundary could be described in a formally rigorous way.

Finally, the proposed framework inevitably raises the question of the relation between its conceptual potential and the demands of modern science for testability. If this model is to claim further development, it will have to generate—or at least inspire—concrete research programs capable of giving new interpretations of already known phenomena or formulating new testable hypotheses.

For this reason, the present article should be treated not as a completed theory, but as a first step: an attempt to outline a new conceptual perspective and to introduce a language that allows questions to be posed beyond the traditional division into object, subject, space, time, and information.

> The proposed framework is not an answer, but an invitation to critical discussion as to whether the world, space, time, and the subject S may be not primary entities, but the result of multilevel compression of the Difference Field D.

X. CONCLUSION

The thought experiment considered in this article, involving a room filled with homogeneous fog, shows that the physical presence of an environment does not yet guarantee the existence of a world for the subject S. Even in an environment saturated with molecules, photons, and potential configurations, the world can almost disappear if the subject S has no access to invariant structures $\mathcal{I}$ sufficient for the construction of a coordinated picture.

On this basis, the article has proposed distinguishing between data, information, and the accessible world. Data encompass everything that is physically present. Information arises only where part of the data becomes accessible in the form of invariant structures $\mathcal{I}$. The accessible world is narrower still: it is a system of invariant structures $\mathcal{I}$ integrated into a stable picture for a particular subject S.

To describe a more fundamental level, the notion of the Difference Field D was introduced—a field that is neither an already given world nor space nor time. Space and time, within the proposed framework, appear as derivative modes of coordinating invariant structures $\mathcal{I}$. This is why they may radically change or almost disappear when the corresponding invariant structures become inaccessible.

The hypothesis of multilevel compression was then advanced,

$$D \to D' \to D'' \to D''' \to \dots$$

within which each new level decreases the quantity of accessible differences while increasing the integrability of invariant structures $\mathcal{I}$. In this perspective, the subject S is not a primary entity but appears as a higher-order invariant structure capable of integrating accessible invariant structures $\mathcal{I}$ into a coordinated world and maintaining its own continuity.

The proposed model makes it possible to look anew at the problem of subjectivity, defining it not as a closed or self-sufficient essence, but as a dynamic process of maintaining invariance within the Difference Field D. In this perspective, the subject S and the world do not precede one another, but exist together as two sides of the same process: the world is a system of invariant structures $\mathcal{I}$ accessible to S, while the subject S is the structure that integrates, retains, and coordinates these invariants.

Although the author treats the framework of the future theory formulated at this stage with due caution, already now, on the basis of the considerations presented, it is possible to attempt to advance new hypotheses and outline promising research directions in such areas as fundamental physics, mathematics, cybernetics, neurobiology and biology in general, gerontology, the search for mind (not only extraterrestrial), and the development of artificial intelligence.

Thus, the main result of this work lies not in the construction of a completed theory, but in the formulation of a new conceptual perspective. It allows the subject S, the world, space, time, and information to be regarded not as independent and primary entities, but as interrelated manifestations of a single process—the process of the emergence, compression, and maintenance of invariance within the Difference Field D.

It is precisely for this purpose that the present work is published: not as a completed system, but as an attempt to outline a language and framework capable of becoming the basis for further critical, formal, and interdisciplinary discussion.

[1] T. Metzinger, *Being No One* (MIT Press, Cambridge, MA, 2003).
[2] G. Bateson, *Steps to an Ecology of Mind* (University of Chicago Press, Chicago, 1972).
[3] K. Friston, Nature Reviews Neuroscience **11**, 127 (2010).
[4] F. J. Varela, E. Thompson, and E. Rosch, *The Embodied Mind* (MIT Press, Cambridge, MA, 1991).
[5] B. B. Mandelbrot, *The Fractal Geometry of Nature* (W. H. Freeman and Company, New York, 1982).
[6] B. B. Mandelbrot, *Fractals: Form, Chance and Dimension* (W. H. Freeman and Company, San Francisco, 1977).
[7] D. O. Hebb, Scientific American (1958).

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